

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Section	02
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Case Study	KADA (Registration and loan application)
System Name	Koperasi Kakitangan KADA SDN.BHD
Group Name	DATA ALCHEMIST

SECTION A (SYSTEM'S OVERVIEW)		
1.1	Overall Description	
		<< <i>Describe the system's overview.</i> >>
1.2	Project Weaknesses or Improvement	
		<< <i>List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses.</i> >>
1.3	Proposed Module	
		<< <i>Describe a new module to overcome the project's weaknesses.</i> >>
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	Conceptual ERD	
		<< <i>Draw a conceptual ERD to represent the proposed module.</i> >>
2.2	Logical ERD	
		<< <i>Provide logical ERD of your module in the proposed system.</i> >>
2.3	Data Dictionary for Logical ERD	
		<< <i>Provide the data dictionary based on the logical ERD in Subsection 2.2.</i> >>
2.4	Relational Database Schema	
		<< <i>Provide all relations database schema of your proposed module.</i> >>
2.5	Normalization	
		<< <i>Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved.</i> >>

1. SECTION A (SYSTEM OVERVIEW)

1.1.Overall Description

Our client is Koperasi Kakitangan KADA. The system that we would propose is the KADA system. The KADA system that we proposed is a web-based system which is different from the manual system. The system mainly functions as members of KADA could apply for a loan through the website only. They don't have to apply it manually to the headquarters as it could save more time and money for the users. In addition, by having this e-system, users also may know their current applications status whether it's pending, approved, or rejected by clicking the "Status Permohonan Pembiayaan" button on the home page. Furthermore, before users would like to apply for a loan, they could also make a rough calculation on the Loan Calculator as they will enter their type of loan, amount of loan, and the duration of repayment, so that they could choose the most beneficial type of loan for them.

To get this idea of the proposed system, our group also did not run from having a minor issue in the beginning of the project. For example, we don't get clear information regarding how the loan approval process and membership process works. But as we are actively engaged with KADA, we could handle this situation and come up with this proposed system. We divided our proposed system into 4 modules. Table 1.1 shows the modules that we have figured out and divide it equally among us.

Group Members' Name	Matric No.	Module
Yasmin Batrisyia Binti Zahiruddin	A23CS0201	Loan Application UC002 - applying loan UC003 - check loan status UC012 - calculate loan payment
Muhammad Naim Bin Abdullah	A23CS0134	Clerk UC009 - create an account UC010 - manage list of member

		UC011 - manage loan application manage list of member
Muhammad Mukhritz Al Iman Bin Mohd Raffi	A23CS0250	ALK UC007 - verify loan application UC008 - view annual report UC013 - Payment
Syarifah Dania Binti Syed Abu Bakar	A23CS0183	User Management UC001 - Manage Profile UC004 - login * UC005 - register * UC006 - Logout

Table 1.1

1.2. Project Weaknesses or Improvement

Based on my module in the WBL project, I have identified a significant flaw in the loan application procedure. Once members have submitted their loan application form, there are no opportunities for them to make any adjustments to it. For example, if a member's guarantor for the loan application unexpectedly passes away during the loan period, the current system does not provide a guarantor replacement form. As a result, members are left with only one guarantor for the loan, which means that it does not fulfill the loan requirement of having two guarantors.

Therefore, the improvement I would like to make based on my module is to introduce a guarantor replacement form for members. This shows that Cooperative KADA is very aware of its members' needs and life changes, and it can also increase member satisfaction. Having this official form also ensures that all changes made are documented and authorized.

1.3. Proposed Module

The new module that I would like to propose is a guarantor replacement module. This module will include two actors: members and clerks. The system will provide a guarantor replacement form on the homepage of the system. If the members of Cooperative KADA would like to make an adjustment regarding their loan guarantor, they can fill in the guarantor replacement form provided and submit it through the system. Then, the form will be sent to the clerk for checking and approval. If the clerk approves the replacement guarantor for the members, it will be updated in the system list and database. The members will then be notified via email that the guarantor replacement has been successfully changed.

2. SECTION B (DATABASE PALNNING AND DESIGN)

2.1. Conceptual ERD

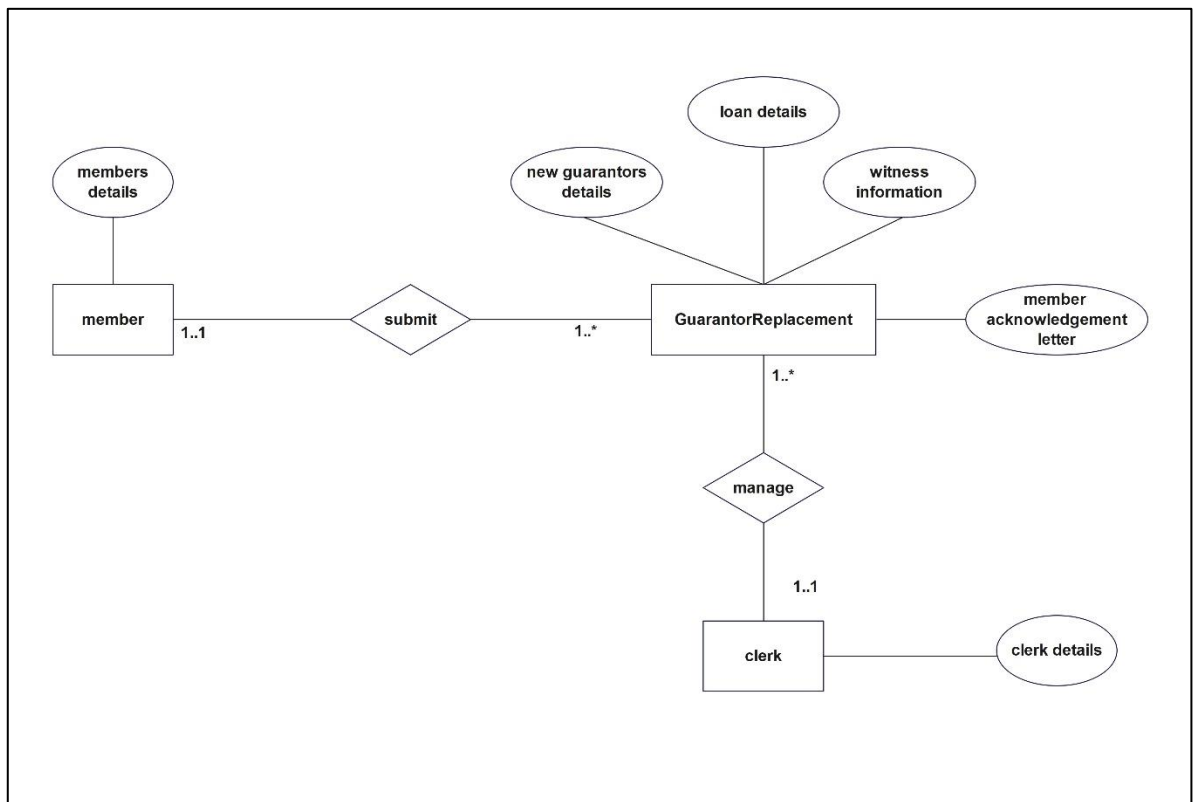


Figure 1

2.2.Logical ERD

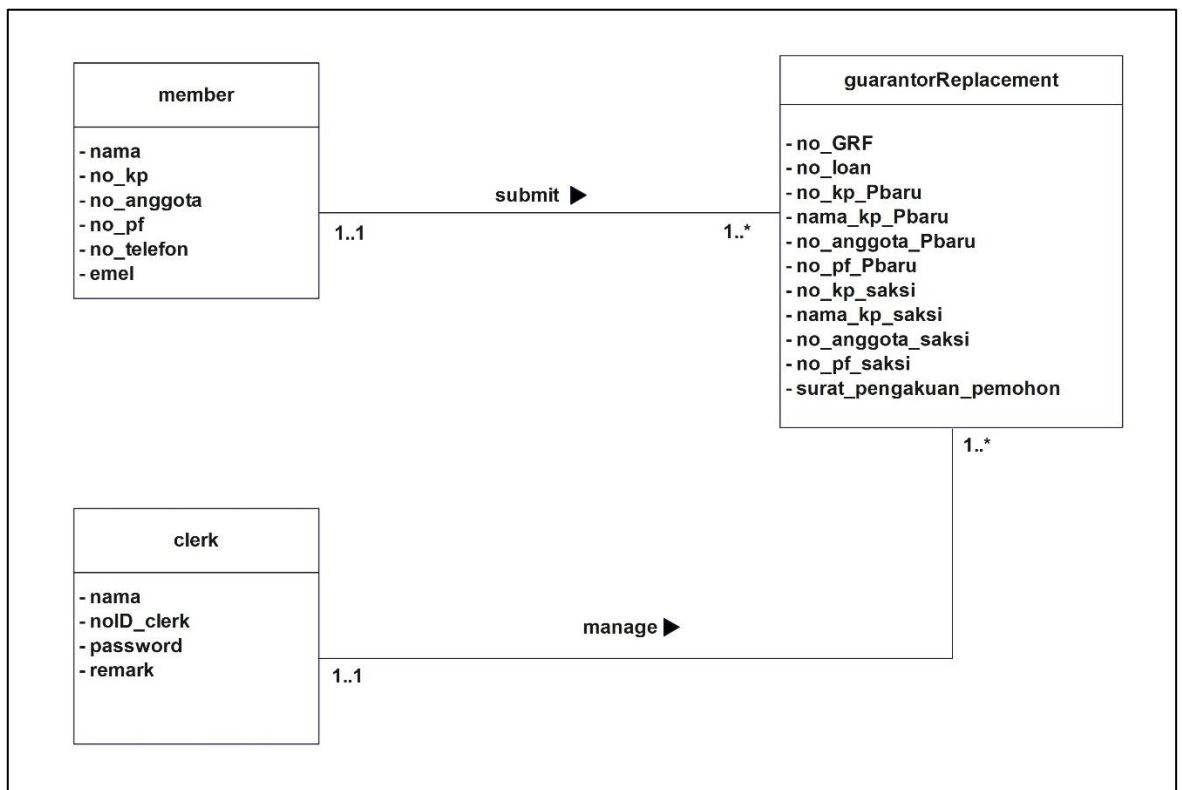


Figure 2

2.3.Data Dictionary for Logical ERD

Entity Name	Attributes	Description	Data type & Length	Null	Multivalued
Member	no_anggota	uniquely identify each membership number	Varchar (255)	no	no
	nama	members name	Varchar (255)	no	no
	no_kp	member identification number	Varchar (255)	no	no
	no_pf	members personal file number	Varchar (255)	no	no
	no_telefon	members phone number	Varchar (255)	no	no
	emel	members email	Varchar (255)	no	no
Clerk	noID_clerk	clerk id number	Varchar (255)	no	no
	nama	clerks full name	Varchar (255)	no	no
	password	clerks passwords	Varchar (255)	no	no
	remark	remarks to accept / reject	Varchar (255)	no	no
GuarantorReplacement	no_GRF	unique number guarantor replacement form	Varchar (255)	no	no
	no_loan	loan registered number	Varchar (255)	no	no
	no_kp_Pbaru	new guarantor ic number	Varchar (255)	no	no
	nama_Pbaru	new guarantor full name	Varchar (255)	no	no
	no_anggota_Pbaru	new guarantor no anggota	Varchar (255)	no	no
	no_pf_Pbaru	new guarantor personal file number	Varchar (255)	no	no
	no_kp_saksi	witness ic number	Varchar (255)	no	no
	nama_saksi	witness full name	Varchar (255)	no	no
	no_anggota_saksi	witness no anggota	Varchar (255)	no	no
	no_pf_saksi	witness personal file number	Varchar (255)	no	no
	statusReplacement	current status replacement	Varchar (255)	no	no
	suratPengakuanPemohon	members upload their letter of acknowledgment	Varchar (255)	no	no

Table 1.2

2.4. Relational Database Schema

1. Member (nama, no_kp, no_anggota, no_pf, no_telefon, emel)

Primary key: no_anggota

2. GuarantorReplacement (no_GRF, no_loan, no_kp_Pbaru, nama_Pbaru, no_anggota_Pbaru, no_pf_Pbaru, no_kp_saksi, nama_saksi, no_anggota_saksi, no_pf_saksi, statusReplacement, suratPengakuanPemohon, noID_clerk, no_anggota)

Primary key: no_GRF

Foreign key: noID_clerk, no_anggota

3. Clerk (noID_clerk, nama, password, remark)

Primary key: noID_clerk

2.5.Normalization

First Normal Form (1NF)

1. Member (nama, no_kp, no_anggota, no_pf, no_telefon, emel)

Primary key: no_anggota

Alternate key: no_pf

FD1: no_anggota -> nama, no_kp, no_pf, no_telefon, emel

FD2: no_pf -> nama, no_kp, no_anggota, no_telefon, emel

2. Clerk (noID_clerk, nama, password, remark)

Primary key: noID_clerk

Alternate key: password

FD1: noID_clerk -> nama, password, remark

FD2: password -> noID_clerk, nama, remark

3. GuarantorReplacement (no_GRF, no_loan, no_kp_Pbaru, nama_Pbaru, no_anggota_Pbaru, no_pf_Pbaru, no_kp_saksi, nama_saksi, no_anggota_saksi, no_pf_saksi, statusReplacement, suratPengakuanPemohon, noID_clerk, no_anggota)

Primary key: no_GRF

Foreign key: noID_clerk, no_anggota

FD1: no_GRF -> no_loan, no_kp_Pbaru, nama_Pbaru, no_anggota_Pbaru, no_pf_Pbaru, no_kp_saksi, nama_saksi, no_anggota_saksi, no_pf_saksi, statusRemplacement, suratPengakuanPemohon, noID_clerk, no_anggota)

Second Normal Form (2NF)

1. Member (nama, no_kp, no_anggota, no_pf, no_telefon, emel)

Primary key: no_anggota

Alternate key: no_pf

FD1: no_anggota -> nama, no_kp, no_pf, no_telefon, emel

FD2: no_pf -> nama, no_kp, no_anggota, no_telefon, emel

2. Clerk (noID_clerk, nama, password, remark)

Primary key: noID_clerk

Alternate key: password

FD1: noID_clerk -> nama, password, remark

FD2: password -> noID_clerk, nama, remark

3. GuarantorReplacement (no_GRF, no_loan, statusReplacement, suratPengakuanPemohon, no_anggota_Pbaru, no_anggota_saksi, noID_clerk, no_anggota)

Primary key: no_GRF

Foreign key: noID_clerk references clerk (noID_clerk), no_anggota references (no_anggota)

4. Guarantor (no_anggota_Pbaru, no_kp_Pbaru, nama_Pbaru, no_pf_Pbaru)

Primary key: no_anggota_Pbaru

5. Witness (no_anggota_saksi, no_kp_saksi, nama_saksi, no_pf_saksi)

Primary key: no_anggota_saksi

Third Normal Form (3NF)

- No 3NF because there is no transitive or dependency attributes in any table.